

5. Computer System Organization & Architecture

Computer Science Fundamentals | Module 2: System Organization | CS_Fundamental-00

Reference to Previous Concepts: Concept 2.1 & 2.3 mein humne Hardware resources (CPU, Memory, I/O) dekhe the. Yeh topic **Image 5** par aadharit hai jisme modern system ka **Common Bus Architecture, Device Controllers**, aur bottom slide mein **CPU Internal Architecture (ALU, CU, Registers, I/O Modules)** explain kiye gaye hain.

5.1 Modern Computer System Operation & Common Bus

- **Structure:** Modern general-purpose computer system consists of one or more CPUs and a number of device controllers connected through a **Common System Bus** that provides access to shared memory.
- **Multicore Architecture (Image 5 Note):** Single chip par multiple execution cores hone se parallel computing aur high throughput milta hai.

5.2 Device Controllers & Concurrent Execution

- **Specialized Controllers:** Disk Controller (Storage), USB Controller (Peripherals), Graphics Adaptor (Video output).
- **Memory Cycle Competition:** CPU aur Device Controllers **concurrently (ek sath)** execute hote hain aur shared memory access ke liye compete karte hain.

5.3 Memory Controller & Synchronization

Orderly access ensure karne ke liye **Memory Controller** provide kiya jata hai jo memory access ko synchronize karta hai aur data clash rokta hai.

Fig 5.1: Computer System Organization & CPU Architecture

Reference: Image 5 [Modern System Bus & Internal CPU Operations]

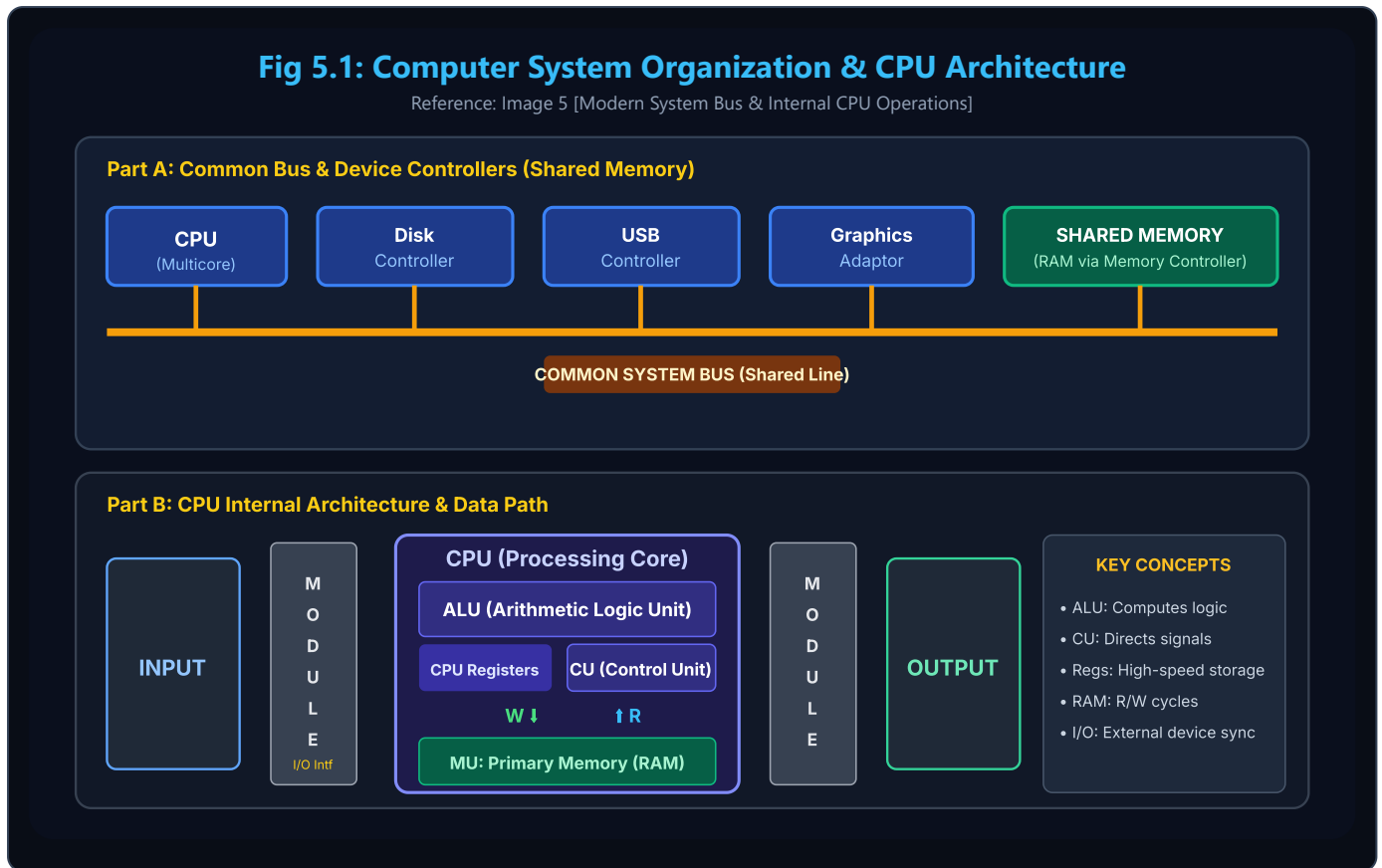


Fig 5.1: Common Bus Architecture & CPU Internal Data Path

5.4 CPU Internal Architecture (Image 5 Bottom Handwritten Diagram)

- **ALU (Arithmetic Logic Unit):** Arithmetic operations (+, -, *, /) aur logical decisions (AND, OR, NOT).
- **CU (Control Unit):** Data timing aur instruction sequencing control karta hai.
- **CPU Registers:** High-speed internal storage for immediate data.
- **Memory Unit (MU / RAM):** Read (R) aur Write (W) operations.
- **I/O Interface Modules:** Input data receive karna aur processed output display/devices tak bhejna.